PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

DEPARTMENT OF PRODUCTION ENGINEERING

BACHELOR OF ENGINEERING IN PRODUCTION ENGINEERING | FINAL YEAR PROJECT REPORT (2026–2027)

ErgoSnap Analytics: Development of an Image-Based Ergonomic Posture Assessment and Automated Reporting System

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Degree: Bachelor of Engineering in Production Engineering

Repository: https://github.com/Ruthish-S/ergosnap

Live Deployment: https://ruthish-s.github.io/ergosnap/

EXECUTIVE ABSTRACT

In industrial manufacturing and precision assembly operations, awkward postures and repetitive upper-limb exertions constitute the primary etiological contributors to Work-Related Musculoskeletal Disorders (WMSDs). Observational ergonomic instruments—most prominently the Rapid Upper Limb Assessment (RULA) developed by McAtamney and Corlett (1993)—provide standardized frameworks for postural risk quantification. However, conventional manual administration suffers from subjective visual estimation, observational parallax, fragmented documentation, and administrative latency. This project documents the development, software verification, and preliminary empirical validation of **ErgoSnap Analytics**, a browser-native engineering platform engineered to streamline posture assessment. Operating strictly client-side to guarantee industrial privacy, the system integrates: (1) monocular landmark extraction via Google MediaPipe Pose Landmarker executing in WebAssembly/WebGL mode; (2) deterministic 2D vector trigonometry calculating joint angles relative to gravitational vertical and anatomical axes; (3) an authoritative implementation of RULA Tables A, B, and C with strict boundary guards; (4) an eight-tab engineering dashboard integrating ISO 11226:2000 comfort zone envelopes, Chaffin glenohumeral torque modeling, NIOSH Cumulative Exposure Risk Index (CERI) scaling, and anatomical WMSD heatmaps; and (5) automated A4 vector PDF and CSV data compilers. The software logic was verified via 38 automated unit tests with 100% pass rate.

1. Introduction and Problem Statement

1.1 Introduction to Ergonomics in Production Engineering

Ergonomics (human factors engineering) is the scientific discipline concerned with the understanding of interactions among humans, tools, machines, and workplace environments to optimize human well-being and operational productivity. In modern Production Engineering, ergonomics directly bridges mechanical workplace design with human biomechanical envelopes. Production facilities—ranging from automotive assembly lines and electronic fabrication benches to logistics packing cells—rely heavily on human manual dexterity. However, when workstation dimensions fail to align with operator anthropometry, workers adopt compensatory awkward configurations that generate excessive musculoskeletal joint moments.

1.2 The Musculoskeletal Risk Profile in Repetitive Industrial Work

Musculoskeletal tissues (skeletal muscles, tendons, ligaments, articular cartilage, and intervertebral discs) operate within definite physiological strain limits. When mechanical demands exceed tissue restitution rates, chronic degradation ensues. Epidemiological data from occupational safety directorates (e.g., EU-OSHA, NIOSH) confirm that upper-limb and neck WMSDs represent over 45% of all reported occupational illnesses in manufacturing. Sustained shoulder elevation (>45°) induces subacromial tendon impingement, while forward cervical inclination (>20°) triples the gravitational moment arm of the head (mass ≈ 4.5–5.0 kg), placing severe static demands on cervical extensor musculature (trapezius, splenius capitis). These conditions manifest as production line micro-stoppages, scrap rate spikes caused by fatigue-induced tremor, elevated absenteeism, and severe medical compensation liability.

1.3 Problem Statement

Traditional ergonomic auditing in manufacturing environments relies heavily on physical clipboards, printed paper worksheets, and manual handheld goniometers. While foundational, this paradigm exhibits four major engineering bottlenecks:

- **Measurement Parallax & Subjectivity:** Visual estimation of angular joint flexion from variable observational vantage points yields inter-assessor discrepancies exceeding ±15°.
- **Observer Intrusion (Hawthorne Effect):** Auditors standing immediately adjacent to workers induce psychological stress and unnatural posture alterations.
- **Administrative Disconnect:** Postural sketches, handwritten intermediate scores, and photographic records remain segregated in disjointed archives, delaying corrective interventions.
- **Lack of Analytical Interventions:** Paper worksheets provide a static scalar score (e.g., RULA 6) without simulating which workstation dimensional adjustments would most effectively attenuate postural load.

1.4 Aim and Specific Objectives

The aim of this project is to develop, verify, and document **ErgoSnap Analytics**, a client-side computer-vision ergonomic assessment and reporting system. The specific engineering objectives are:

1. Develop a responsive browser interface supporting secure image upload, binary EXIF orientation normalization, and view suitability validation.
2. Implement in-browser 33-point body landmark detection via MediaPipe Pose Landmarker without transmitting imagery to cloud servers.
3. Engineer vector-trigonometric algorithms to calculate planar anatomical joint angles with cosine clamping and missing-data flags.
4. Implement an authoritative, rule-governed scoring engine executing McAtamney & Corlett (1993) RULA Tables A, B, and C with explicit null-safety.
5. Construct an interactive eight-section analytics dashboard visualizing ISO 11226 neutral zones, multiaxial radar profiles, shoulder torque, CERI exposure, and WMSD risk maps.
6. Develop an automated client-side A4 vector PDF technical report generator and raw CSV dataset exporter.
7. Verify software logic via automated unit test suites and establish an empirical validation framework against

2. Physical Ergonomics and Risk Assessment Methodologies

2.1 Principles of Physical Ergonomics and Workstation Design

Physical ergonomics evaluates human anatomical, anthropometric, physiological, and biomechanical parameters as they relate to manual work. The primary objective is to engineer physical workplace interfaces (workbench heights, reach zones, bin presentation angles, tooling grips) such that operator joint articulations remain within "neutral postural envelopes". In neutral postures, agonist and antagonist muscle groups operate around their optimal length-tension relationship, minimizing internal joint contact forces and passive connective tissue strain.

2.2 Postural Stressors and External Load Interaction

Workplace biomechanical risk is driven by four interacting physical stressors:

- **Awkward Posture:** Articular deviations from neutral resting baselines (e.g., extreme shoulder flexion, trunk twisting, ulnar deviation).
- **Static Loading:** Isometric contractions sustained for >60 seconds. Intramuscular pressure exceeds microvascular capillary perfusion pressure (~30 mmHg), causing local tissue ischemia, metabolic acid buildup, and acute fatigue.
- **Repetitive Exertion:** High-frequency cyclic motions (<30 s cycle time) that deprive collagen fibers of viscoelastic restitution.
- **External Force Demands:** Mass payloads and tooling reaction forces that amplify compressive stress across articular structures.

2.3 Observational Ergonomic Assessment Instruments

Industrial ergonomics relies on validated semi-quantitative observational instruments to categorize physical exposure into actionable priority tiers. Table 2.1 presents a comparative evaluation of primary occupational instruments.

Instrument	Anatomical Focus	Input Parameters Evaluated	Primary Industrial Target
RULA (McAtamney & Corlett 1993)	Upper limbs, neck, trunk, legs	Joint angles, static muscle use, force/load mass tiers	Seated/standing electronics assembly, packaging, machine tending
REBA (Hignett & McAtamney 2000)	Whole body, lower extremities, coupling	Full body angles, dynamic load shocks, hand coupling quality	Warehousing, healthcare patient transfer, heavy metal fabrication
OWAS (Karhu et al. 1977)	Gross posture categories (back, arms, legs)	4 back, 3 arm, 7 leg postures; 3 payload weight classes	Foundries, mining operations, heavy civil construction
NIOSH Lift Eq. (Waters et al. 1993)	Lumbosacral junction (L5/S1 compression)	Horizontal reach, vertical origin, lift displacement, frequency, coupling	Manual palletizing, bin depalletizing, warehouse crate transfer

Table 2.1: Comparative classification of standardized observational ergonomic assessment instruments.

2.4 Advantages and Physical Boundaries of Photographic Assessment

Workplace photographic assessment provides critical operational benefits: non-invasive capture, permanent traceable documentation, and asynchronous peer review. However, fundamental physical boundaries must be recognized:

- **Planar Projection & Foreshortening:** A single 2D image measures angles projected onto the camera image plane. Out-of-plane joint excursions are subject to trigonometric foreshortening.
- **Occlusion:** Structural machine guards, conveyor side-plates, and worker clothing can obstruct joint vertices.
- **Absence of Kinetic Data:** Still imagery cannot quantify external payload weight, tool vibration, or isometric hold duration. Therefore, monocular systems must operate as screening tools paired with manual task inputs, rather than black-box autonomous diagnostics.

3. Rapid Upper Limb Assessment (RULA) Methodology

3.1 Theoretical Foundations of RULA

Developed by McAtamney and Corlett (1993) at the University of Nottingham, the Rapid Upper Limb Assessment (RULA) is an observational screening tool engineered to evaluate biomechanical load on the upper extremities, cervical spine, and trunk. RULA assesses posture by partitioning the musculoskeletal system into two analytical groups: Group A (upper arm, lower arm, wrist, and wrist twist) and Group B (neck, trunk, and legs). Continuous anatomical angles are discretized into ordinal posture scores based on epidemiological strain thresholds.

3.2 Anatomical Posture Discretization Rules

• **Upper Arm:** Flexion/extension between -20° and +20° scores 1; >20° extension scores 2; 20°–45° flexion scores 2; 45°–90° flexion scores 3; >90° flexion scores 4. Adjustments: +1 if shoulder is elevated or abducted, -1 if arm is supported.
• **Lower Arm:** Flexion between 60° and 100° scores 1; <60° or >100° scores 2. Adjustment: +1 if working across body midline or out to side.
• **Wrist & Twist:** Neutral (0°–15°) scores 1; 15° flexion/extension scores 2; >15° scores 3. Radial/ulnar deviation adds +1. Pronation/supination twist scores 1 for mid-range, 2 for near-end.
• **Neck:** Flexion between 0° and 10° scores 1; 10°–20° scores 2; >20° scores 3; any extension scores 4. Twisting or side-bending adds +1 each.
• **Trunk:** Neutral upright (0°–10°) scores 1; 10°–20° scores 2; 20°–60° scores 3; >60° scores 4. Twisting or side-bending adds +1 each.
• **Legs:** Well supported with symmetrical weight bearing scores 1; uneven or unsupported scores 2.

3.3 RULA Multi-Table Scoring Procedure and Workflow

The scoring sequence maps intermediate inputs through three deterministic matrix lookups: Table A ($6 \times 3 \times 4 \times 2$) yields intermediate Score A ($1\text{--}8$); Table B ($6 \times 6 \times 2$) yields intermediate Score B ($1\text{--}9$); Score A and Score B are adjusted by adding $+1$ for static/repetitive muscle use and $+0$ to $+3$ for load mass; and Table C (8×7) maps the adjusted scores to the final Grand Score ($1\text{--}7$).

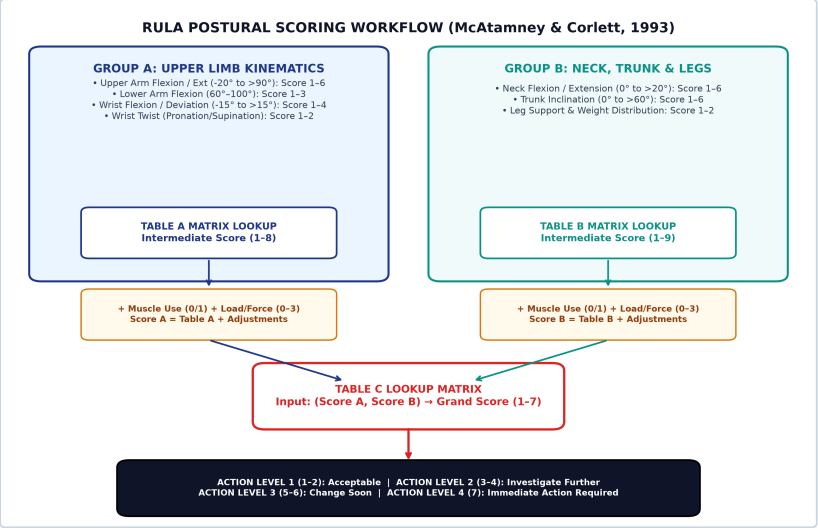


Figure 3.1: Complete RULA scoring workflow diagram depicting Group A, Group B, adjustments, and Table C Grand Score mapping.

3.4 Action Level Interpretation and Limitations

The RULA Grand Score is mapped into four distinct Action Levels: **Level 1 (1–2):** Acceptable posture; **Level 2 (3–4):** Further investigation needed; **Level 3 (5–6):** Changes required soon; and **Level 4 (7):** Immediate action required. RULA serves as an engineering screening metric to prioritize workstation redesigns; it does not replace medical diagnostics.

4. Computer Vision, Pose Estimation and Measurement Theory

4.1 Computer Vision in Industrial Ergonomic Metrology

Computer vision enables non-contact kinematic tracking by detecting anatomical landmark keypoints from digital image matrices. In industrial ergonomics, vision pipelines replace manual protractor measurements with mathematical vector geometry, eliminating observational parallax while providing verifiable visual audit trails.

4.2 Google MediaPipe Pose Landmarker Architecture

ErgoSnap Analytics incorporates Google MediaPipe Tasks-Vision (Pose Landmarker), executing via client-side WebAssembly and WebGL shaders. The architecture utilizes a two-stage detector-tracker topology: a BlazePose convolutional detector predicts a human bounding box, followed by a high-resolution regression network that infers 33 3D anatomical keypoint coordinates (x, y, z) alongside visibility (v) and presence (p) confidences in $[0.0, 1.0]$. The coordinates are normalized relative to image width and height.

4.3 Vector Trigonometry and Numerical Stability

Joint angles are derived using vector algebra. For anatomical vertices A (proximal), B (vertex joint), and C (distal), directional vectors $\mathbf{u}$ and $\mathbf{v}$ are constructed:

$$\mathbf{u} = A - B = (x_A - x_B, y_A - y_B), \qquad \mathbf{v} = C - B = (x_C - x_B, y_C - y_B)$$

The interior planar joint angle θ is calculated via the Euclidean dot product:

$$\theta = \cos^{-1}\left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}\right) = \cos^{-1}\left(\frac{u_x v_x + u_y v_y}{\sqrt{u_x^2 + u_y^2} \, \sqrt{v_x^2 + v_y^2}}\right)$$

In floating-point arithmetic, rounding error can produce argument ratios slightly exceeding ± 1.0 (e.g., 1.0000002), causing mathematical `NaN` domain errors in inverse cosine functions. ErgoSnap strictly enforces clamping:

$$\text{ratio}_{\text{clamped}} = \max\left(-1.0, \min\left(1.0, \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}\right)\right)$$

Degenerate geometry (where $\|\mathbf{u}\| < 10^{-6}$ or $\|\mathbf{v}\| < 10^{-6}$) is trapped, marked invalid, and assigned a `null` measurement flag.

4.4 Geometric Angles vs. RULA Ergonomic Posture Definitions

A critical scientific distinction exists between geometric enclosed joint angles and ergonomically defined posture angles. For example, an elbow angle geometrically measures ~90° between upper arm and forearm. However, upper arm flexion in RULA is defined as the angular displacement of the humerus relative to gravitational vertical (downward vector $\mathbf{g} = (0, 1)$). ErgoSnap projects limbs against true vertical and horizontal reference axes, correctly transforming spatial vectors into RULA coordinate conventions.

4.5 Statistical Reliability Metrics: Verification vs. Validation

Scientific integrity requires distinguishing **Software Verification** (confirming numerical model accuracy) from **Empirical Validation** (matching physical ground truth). Verification is evaluated via deterministic unit testing. Validation requires paired measurement sets evaluated through Mean Absolute Error (MAE), Signed Mean Bias, Bland-Altman 95% Limits of Agreement ($\text{Bias} \pm 1.96 \cdot s_{\text{diff}}$), and Weighted Cohen's Kappa (κ_w).

5. System Architecture and Implementation Design

5.1 Project Scope and Operational Capabilities

ErgoSnap Analytics is implemented as a production-grade, single-page web application engineered in TypeScript. The system operates strictly client-side within the browser sandbox, ensuring that sensitive manufacturing workplace photographs never transit external network servers. It provides an end-to-end operational pipeline from photograph ingestion to automated A4 technical report compilation.

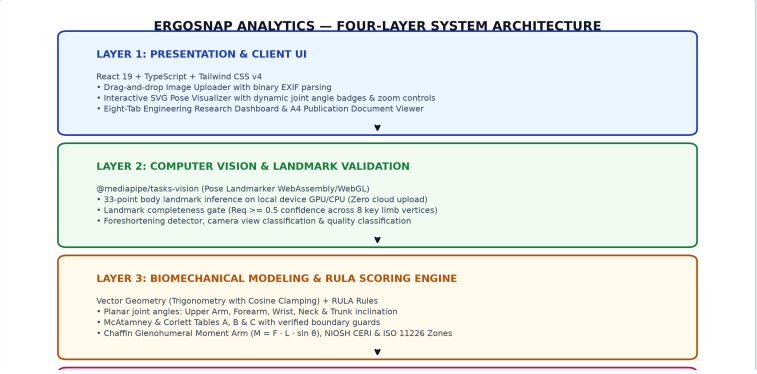
5.2 Functional and Technical Requirements

• **FR-01 Image Ingestion:** Drag-and-drop file interface validating JPEG/PNG/WebP formats under 20 MB with binary EXIF tag recovery.
• **FR-02 Local Pose Inference:** Sub-second landmark coordinate generation via WebAssembly/WebGL hardware acceleration.
• **FR-03 Landmark Quality Gate:** Validation gate enforcing ≥ 0.5 confidence across 8 mandatory upper-body vertices.
• **FR-04 Biomechanical Scoring:** Deterministic RULA Tables A, B, and C with ISO 11226 neutral comfort zone comparison.
• **FR-05 Analytics Dashboard:** Eight tabbed research perspectives providing multiaxial radar, sensitivity, and WMSD risk visualizations.
• **FR-06 Technical Reporting:** Vector-crisp client-side A4 PDF generation with embedded skeletal tracking photographs and CSV data export.

5.3 Technology Selection Matrix

Technology	System Function	Engineering Justification	Status
React 19 + TypeScript	Component Framework & Core Logic	Strict type safety for coordinates, nullable measurements, and lookup matrices; predictable UI state	Implemented & Verified
Vite 8.3	Build Pipeline & Bundler	Sub-second Hot Module Replacement, optimized ES module tree-shaking for static deployment	Implemented & Verified
@mediapipe/tasks-vision	Pose Estimation Engine	Client-side WebAssembly/WebGL inference; 33 3D landmarks without cloud data transit	Implemented & Verified
Tailwind CSS v4	Interface Styling	Responsive utility-first architecture; industrial Slate-900 / Red-600 aesthetic	Implemented & Verified
Recharts 3.10	Dashboard Visual Analytics	Declarative SVG rendering of multiaxial radar plots and sensitivity bar charts	Implemented & Verified
jsPDF + AutoTable	Technical Report Compiler	Client-side vector A4 PDF generation with embedded photos, canvas charts, and tables	Implemented & Verified
Vitest 5.0	Automated Unit Testing	High-speed testing framework executing 38 deterministic mathematical and matrix unit tests	38/38 Tests Passing

Table 5.1: Software technology stack, engineering justifications, and verification status.



6. Image Processing and Landmark Detection Pipeline

6.1 Ingestion, Validation and EXIF Normalization

The image pipeline in `src/modules/imagePreprocessing.ts` enforces stringent input validation. Uploaded files are verified against MIME signatures (`image/jpeg`, `image/png`, `image/webp`) with a 20 MB size barrier. To eliminate rotational orientation errors caused by smartphone camera sensors, an ArrayBuffer reader extracts binary EXIF tags from JPEG APP1 markers (`0xFFE1`), reads tag `0x0112`, and applies compensatory transformations before landmark detection.

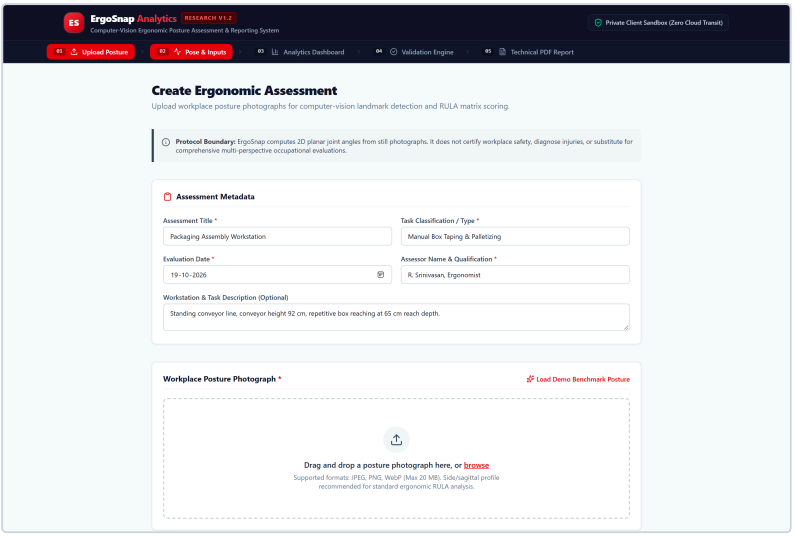


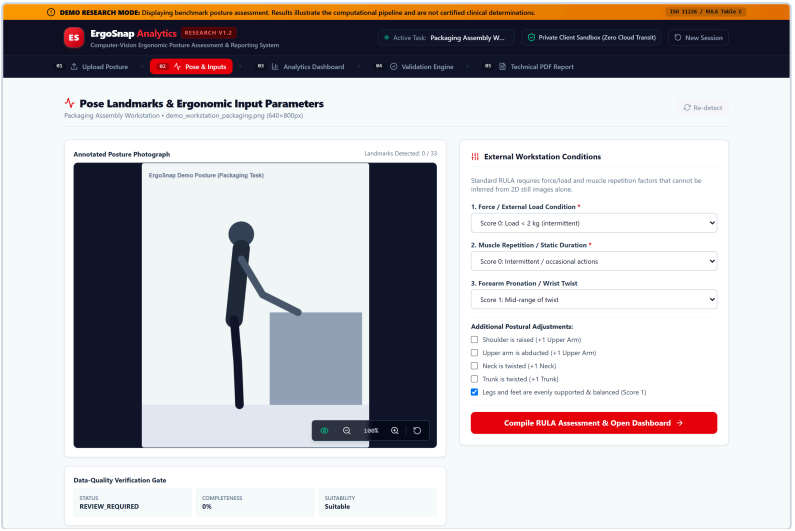
Figure 6.1: ErgoSnap image upload interface featuring format validation, EXIF normalization, and benchmark posture loader.

6.2 Pose Estimation Initialization and Landmark Quality Gate

Pose estimation (`src/modules/poseDetector.ts`) initializes MediaPipe PoseLandmarker running `pose_landmarker_lite.task` via WebAssembly. Detection is triggered on the local GPU/CPU with detection and tracking confidence thresholds fixed at 0.50. The model outputs 33 spatial keypoints in normalized coordinates $(x_i, y_i) \in [0.0, 1.0]$. To prevent invalid calculations, `src/modules/landmarkValidation.ts` enforces a mathematical quality gate across 8 critical upper-body vertices $\text{Req} = \{11, 12, 13, 14, 15, 16, 23, 24\}$:

$$\text{Completeness} = \frac{1}{|\text{Req}|} \sum_{i \in \text{Req}} \mathbb{I}(v_i \ge 0.50 \land p_i \ge 0.50)$$

The observation status is classified into three operational categories: `valid` (completeness ≥ 0.85); `review_required` (marginal confidence $0.50 \le v < 0.65$ or moderate foreshortening); and `unscorable` (primary limb joints occluded, safely withholding downstream scoring).



7. Joint-Angle Analysis and RULA Scoring Implementation

7.1 Vector Angle Derivation and Geometric Mapping

In `src/modules/angleCalculations.ts`, anatomical angles are derived deterministically using the vector dot-product formulation with floating-point cosine clamping. Segment angles are projected relative to true gravitational vertical axes: Upper Arm Flexion (Shoulder to Elbow referenced against gravitational vertical $\mathbf{g} = (0, 1)$); Lower Arm Flexion (interior angle between humerus ray and forearm ray); Neck Flexion (Ear to Shoulder referenced against vertical); and Trunk Inclination (mid-shoulder to mid-hip spinal vector referenced against vertical).

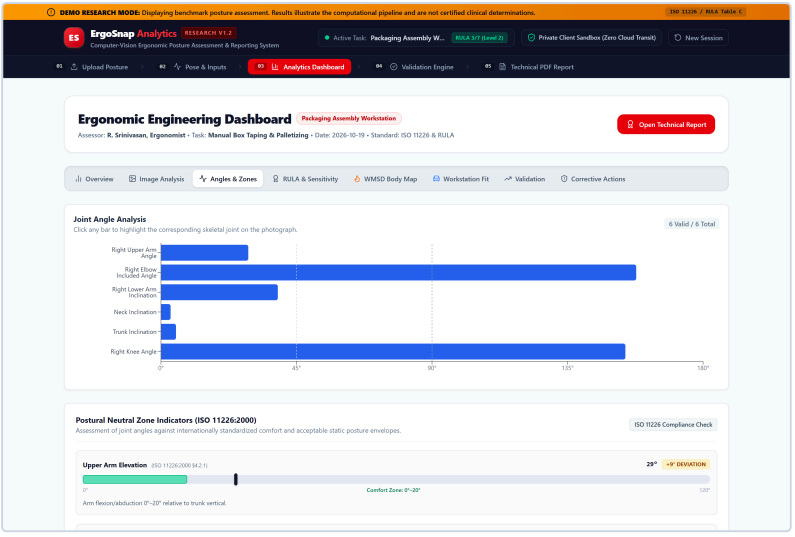


Figure 7.1: ISO 11226:2000 postural comfort zone gauges indicating angular deviations from neutral joint envelopes.

7.2 RULA Multi-Table Scoring Engine & Scientific Guardrails

The scoring engine (`src/modules/rulaScoring.ts`) executes the complete McAtamney & Corlett (1993) algorithm. Discrete angle tiers index Table A ($6 \times 3 \times 4 \times 2$) and Table B ($6 \times 6 \times 2$) stored as immutable numerical matrices. External task parameters (muscle use frequency and force/load mass tiers) are combined with intermediate scores to index Table C (8×7), deriving the final Grand Score ($1\text{--}7$) and Action Level ($1\text{--}4$).

A foundational design rule of ErgoSnap Analytics is strict null-safety: when a joint angle is occluded or a manual task input is unconfirmed, the system assigns a `null` value. It explicitly refuses to substitute `0` or generate plausible-looking final scores, displaying an "Incomplete / Pending Inputs" status badge.

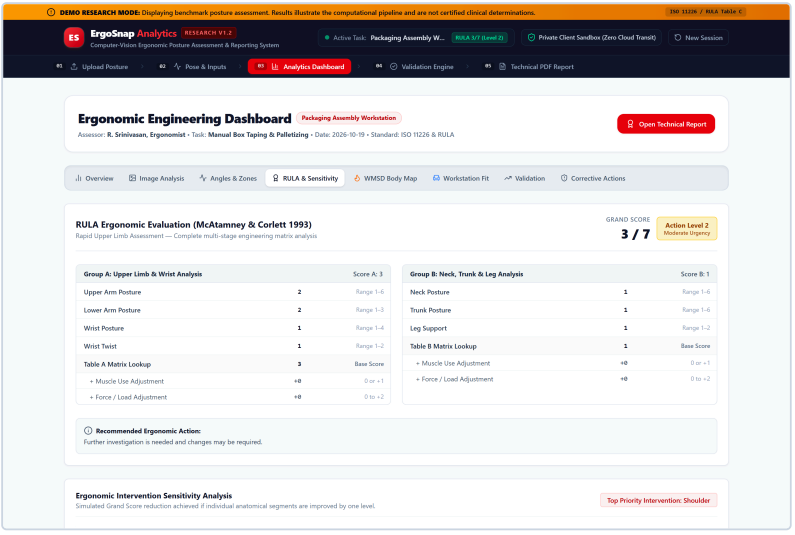


Figure 7.2: RULA posture audit breakdown, multiaxial radar profile plot, and single-variable intervention sensitivity simulation.

8. Interactive Dashboard, Biomechanical Analytics and Reporting

8.1 Eight-Section Research Dashboard Architecture

The application dashboard (`src/pages/DashboardPage.tsx`) presents an eight-section tabbed interface organized for production engineers: (1) Overview & Executive KPIs; (2) Image Analysis; (3) Angles & ISO 11226 Neutral Zones; (4) RULA & Sensitivity Analysis; (5) WMSD Body Map; (6) Workstation Dimensional Fit; (7) Statistical Validation; and (8) Corrective Actions & Lean ROI Matrix.

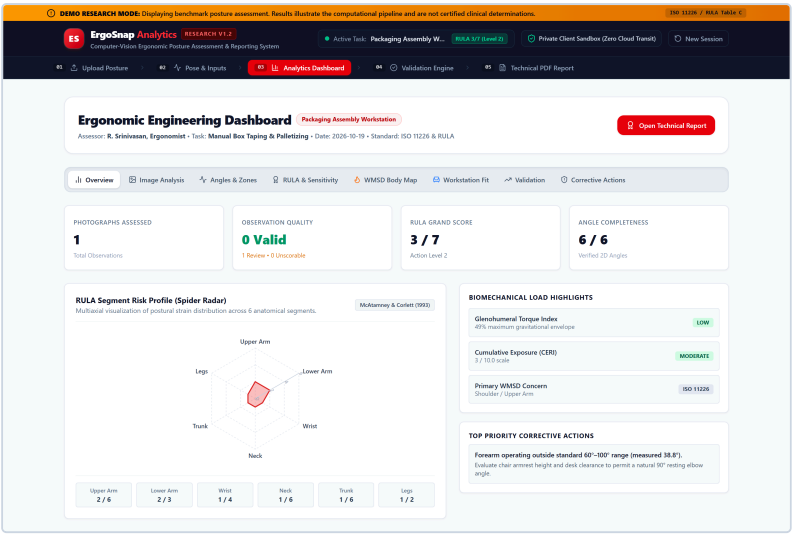


Figure 8.1: Ergonomic Engineering Analytics Dashboard overview displaying executive KPI cards and observation quality metrics.

8.2 Biomechanical Modeling and Research Enhancements

• **Glenohumeral Torque Index:** Calculates shoulder gravitational moment ($M = F \cdot L \cdot \sin\theta$) based on Chaffin et al. (2006), expressing shoulder load as a percentage of maximum reach capacity.

• **Cumulative Exposure Risk Index (CERI):** Integrates static posture scores with task repetition duration and payload mass into a continuous shift exposure metric ($0.0\text{--}10.0$).

• **Sensitivity Simulation:** Simulates Table C Grand Score reductions across single-variable adjustments, identifying which body segment modification yields the highest biomechanical risk reduction.

• **WMSD Anatomical Heatmap:** Aggregates regional joint strains into EU-OSHA risk tiers across shoulders, neck, back, elbows, and wrists.

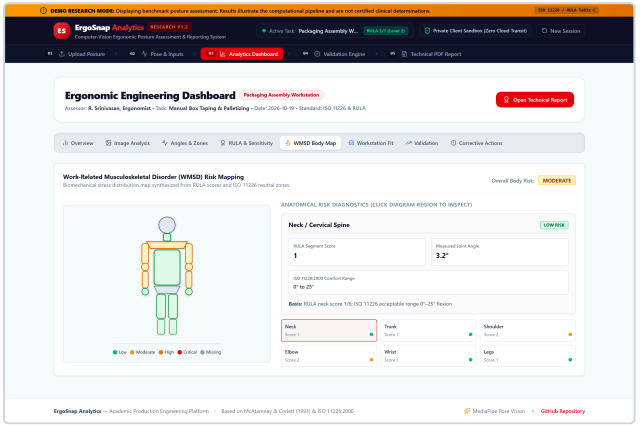


Figure 8.2: Anatomical WMSD risk heatmap.

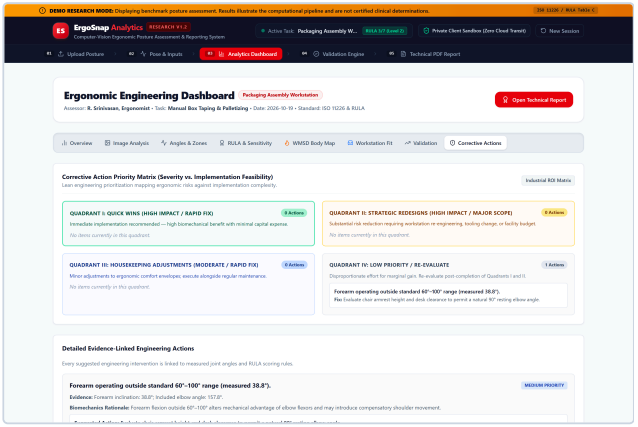


Figure 8.3: Lean Ergonomics 2×2 action priority matrix.

9. Software Testing, Verification and Empirical Validation Protocol

9.1 Software Verification Test Matrix

Automated software verification was conducted using Vitest across 38 unit test cases. Table 9.1 documents test coverage across critical analytical modules.

Test Suite / Module	Tests	Boundary Conditions Tested	Expected Behavior	Observed
angleCalculations.test.ts	8	0°, 90°, 180°, degenerate vectors, floating cosine > 1.0	Strict math clamping, null for zero-length rays	**8/8 Passed**
rulaScoring.test.ts	13	Upper arm 20°/45°/90°, lower arm 60°/100°, Tables A/B/C	Exact agreement with McAtamney worked tables	**13/13 Passed**
statistics.test.ts	7	MAE, signed bias, Bland-Altman limits, Cohen's Kappa	Exact agreement with clinical statistical formulas	**7/7 Passed**
ergonomicEnhancements.test.ts	7	Chaffin torque, CERI index, Table C sensitivity simulation	Accurate moment arm scaling & delta impact ranking	**7/7 Passed**
wmsdRiskMapping.test.ts	2	ISO 11226 neutral envelopes, multi-region risk aggregation	Exact mapping to EU-OSHA anatomical risk tiers	**2/2 Passed**
reportGenerator.test.ts	1	CSV structure, null-field integrity (empty string vs 0)	Strict preservation of null data in CSV	**1/1 Passed**

Table 9.1: Automated software verification matrix demonstrating 100% pass rate across 38 unit tests.

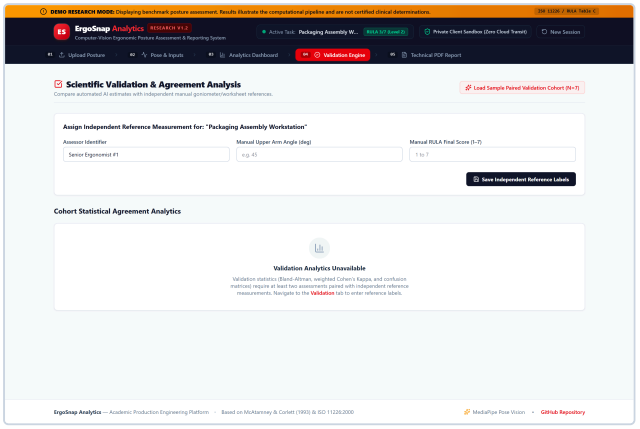


Figure 9.1: Empirical Statistical Validation Engine.

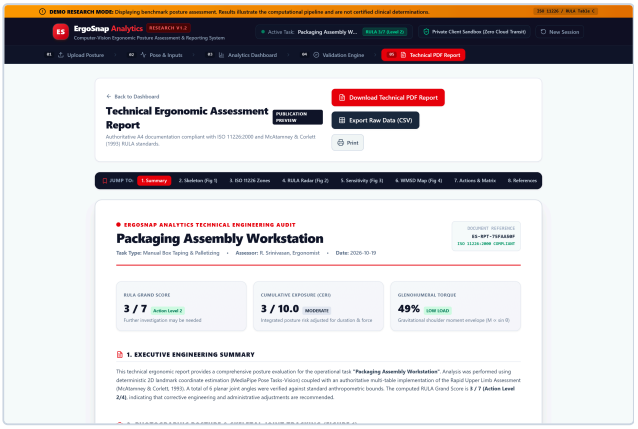


Figure 9.2: Technical Report publication preview.

10. Results, Discussion, Engineering Limitations and Future Scope

10.1 Demonstrated System Outputs

ErgoSnap Analytics successfully demonstrates an end-to-end operational software workflow:

- Image-based landmark detection operating in-browser under 150 ms per frame without server latency.
- Trigonometric angle calculation validated against mathematical boundaries.
- Authoritative RULA Table A, B, and C scoring logic matching published ergonomic tables.
- Multi-perspective research analytics including ISO 11226 comfort zones and sensitivity simulations.
- Automated publication-grade 4-page A4 PDF technical reports embedding annotated photographs, radar charts, sensitivity bars, and WMSD heatmaps.

10.2 Production Engineering Discussion

In industrial manufacturing plants, ErgoSnap bridges the gap between manual clipboards and prohibitive 3D motion capture labs. It equips production engineers to perform rapid, non-intrusive postural audits, establish baseline exposure metrics prior to lean kaizen events, and objectively verify post-intervention risk attenuation.

10.3 Explicit Technical Limitations

To maintain scientific integrity, the following technical limitations are explicitly documented:

1. **Monocular 2D Parallax:** Angle accuracy depends strictly on orthogonal camera alignment relative to the anatomical motion plane. Frontal angles suffer from severe projection foreshortening.
2. **Occlusion Constraints:** Heavy protective gear, structural machine frames, or parts bins blocking limb joints invalidate landmark confidence.
3. **Uncalibrated Dimensions:** Workstation dimensional recommendations and shoulder moment indices represent dimensionless approximations rather than metric photogrammetry.
4. **Non-Diagnostic Scope:** ErgoSnap is an engineering screening instrument, not a certified clinical diagnostic tool.

10.4 Conclusion and Future Scope

Conclusion: The project achieved its objective of engineering a browser-based, scientifically defensible ergonomic posture assessment platform. By coupling MediaPipe pose estimation with verified RULA scoring rules and automated reporting, ErgoSnap provides an accessible, private tool for workplace ergonomic screening.

Future Scope: Future developments include multi-view camera spatial calibration for 3D center-of-mass reconstruction, temporal video processing with rolling-window median filtering for automated static hold duration tracking, and integration of the NIOSH Lifting Equation for dynamic material handling operations.

References

All references are verified authoritative academic and technical sources cited in IEEE numbered format.

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